

From Time Dilation to Mass Variation:

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## Abstract

This updated work presents the essential mathematical formulations of time-mass duality theory, building upon the comprehensive geometric foundations established in the field-theoretic derivation of the  $\beta$  parameter. The theory establishes a duality between two complementary descriptions of reality: the standard view with time dilation and constant rest mass, and the T0 model with absolute time and variable mass. Central to this framework

is the intrinsic time field  $T(x, t) = \frac{1}{\max(m, \omega)}$  (in natural units where  $\hbar = c = \alpha_{\text{EM}} = \beta_{\text{T}} = 1$ ), which enables a unified treatment of massive particles and photons through the three fundamental field geometries: localized spherical, localized non-spherical, and infinite homogeneous. The mathematical formulations include complete Lagrangian densities with a dimensional check, incorporating the derived parameters  $\beta = 2Gm/r$ ,  $\xi = 2\sqrt{G} \cdot m$ , and the cosmic screening factor  $\xi_{\text{eff}} = \xi/2$  for infinite fields. The equations contain no adjustable parameters; not dimensionally uniform are the time field Lagrangian density in the given form and the modified Schrödinger equation with dimensionful  $T$ , which only becomes consistent with the ratio  $T/T_0$ .

## .1 Introduction: Updated T0 Model Foundations

This updated mathematical formulation builds upon the comprehensive field-theoretic foundation established in the T0 model reference framework. The time-mass duality theory now incorporates the complete geometric derivations and natural units system that demonstrate the fundamental unity of quantum and gravitational phenomena.

### Fundamental Postulate: Intrinsic Time Field

The T0 model is based on the fundamental relationship between time and mass expressed through the intrinsic time field:

$$T(x, t) = \frac{1}{\max(m(x, t), \omega)} \quad (1)$$

**Dimensional verification:**  $[T(x, t)] = [1/E] = [E^{-1}]$  in natural units ✓

This field satisfies the fundamental field equation derived from geometric principles:

$$\nabla^2 m(x, t) = 4\pi G \rho(x, t) \cdot m(x, t) \quad (2)$$

**Dimensional verification:**  $[\nabla^2 m] = [E^2][E] = [E^3]$  and  $[4\pi G \rho m] = [1][E^{-2}][E^4][E] = [E^3]$  ✓

### Three Fundamental Field Geometries

The complete T0 framework recognizes three distinct field geometries with specific parameter modifications:

### T0 Model Parameter Framework

#### Localized Spherical Fields:

$$\beta = \frac{2Gm}{r} \quad [1] \quad (3)$$

$$\xi = 2\sqrt{G} \cdot m \quad [1] \quad (4)$$

$$T(r) = \frac{1}{m_0}(1 - \beta) \quad (5)$$

#### Localized Non-spherical Fields:

$$\beta_{ij} = \frac{r_{0ij}}{r} \quad (\text{tensor}) \quad (6)$$

$$\xi_{ij} = 2\sqrt{G} \cdot I_{ij} \quad (\text{inertia tensor}) \quad (7)$$

#### Infinite Homogeneous Fields:

$$\nabla^2 m = 4\pi G \rho_0 m + \Lambda_T m \quad (8)$$

$$\xi_{\text{eff}} = \sqrt{G} \cdot m = \frac{\xi}{2} \quad (\text{cosmic screening}) \quad (9)$$

$$\Lambda_T = -4\pi G \rho_0 \quad (10)$$

### Practical Simplification Note

**For practical applications:** Since all measurements in our finite, observable universe are performed locally, only the **localized spherical field geometry** (first case above) is required:

$\xi = 2\sqrt{G} \cdot m$  and  $\beta = \frac{2Gm}{r}$  for all applications.

The other geometries are shown for theoretical completeness but are not needed for experimental predictions.

## Natural Units Framework Integration

The complete natural units system where  $\hbar = c = \alpha_{\text{EM}} = \beta_T = 1$  provides:

- Universal energy dimensions: All quantities expressed as powers of  $[E]$
- Unified coupling constants:  $\alpha_{\text{EM}} = \beta_T = 1$  as a setting, consistent with Higgs physics to about 2.4 % (Section .5)
- Connection to Planck scale:  $\ell_P = \sqrt{G}$  and  $\xi = r_0/\ell_P$
- Fixed parameter relationships: No adjustable constants in the theory

## .2 Complete Field Equation Framework

### Spherically Symmetric Solutions

For a point mass source  $\rho = m\delta^3(\vec{r})$ , the complete geometric solution is:

$$m(x,t)(r) = m_0 \left( 1 + \frac{2Gm}{r} \right) = m_0(1 + \beta) \quad (11)$$

Therefore:

$$T(r) = \frac{1}{m(x,t)(r)} = \frac{1}{m_0}(1 + \beta)^{-1} \approx \frac{1}{m_0}(1 - \beta) \quad (12)$$

**Geometric interpretation:** The factor 2 in  $r_0 = 2Gm$  emerges from the relativistic field structure, exactly matching the Schwarzschild radius.

### Modified Field Equation for Infinite Systems

For infinite, homogeneous fields, the field equation requires modification:

$$\nabla^2 m(x, t) = 4\pi G \rho_0 m(x, t) + \Lambda_T m(x, t) \quad (13)$$

where the consistency condition for homogeneous background gives:

$$\Lambda_T = -4\pi G \rho_0 \quad (14)$$

**Dimensional verification:**  $[\Lambda_T] = [4\pi G \rho_0] = [1][E^{-2}][E^4] = [E^2] \checkmark$

This modification leads to the cosmic screening effect:  $\xi_{\text{eff}} = \xi/2$ .

## .3 Lagrangian Formulation and Dimensional Check

### Time Field Lagrangian Density

The fundamental Lagrangian density for the intrinsic time field is:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[ \frac{1}{2} g^{\mu\nu} \partial_\mu T(x, t) \partial_\nu T(x, t) - V(T(x, t)) \right] \quad (15)$$

**Dimensional verification:**

- $[\sqrt{-g}] = [E^{-4}]$  (4D volume element)
- $[g^{\mu\nu}] = [E^2]$  (inverse metric)
- $[\partial_\mu T(x, t)] = [E][E^{-1}] = [1]$  (dimensionless gradient)
- $[g^{\mu\nu} \partial_\mu T(x, t) \partial_\nu T(x, t)] = [E^2][1][1] = [E^2]$
- $[V(T(x, t))] = [E^4]$  (potential energy density)
- Total: kinetic term  $[E^{-4}][E^2] = [E^{-2}]$ , potential term  $[E^{-4}][E^4] = [E^0]$

The two terms have different dimensions; with these assignments  $\mathcal{L}_{\text{time}}$  is not dimensionally uniform.

### Modified Schrödinger Equation

The quantum mechanical evolution equation becomes:

$$iT(x, t) \frac{\partial}{\partial t} \Psi + i\Psi \left[ \frac{\partial T(x, t)}{\partial t} + \vec{v} \cdot \nabla T(x, t) \right] = \hat{H} \Psi \quad (16)$$

**Dimensional verification:**

- $[iT(x, t) \partial_t \Psi] = [E^{-1}][E][\Psi] = [\Psi]$
- $[i\Psi \partial_t T(x, t)] = [\Psi][E^{-1}][E] = [\Psi]$
- $[\hat{H} \Psi] = [E][\Psi]$

With a dimensionful  $T = 1/m$  the equation in this form is inhomogeneous: the time-field term carries a different energy dimension than  $\hat{H}\Psi$ . It becomes consistent with the dimensionless ratio  $T/T_0$  in front of the time derivative (A142): for a state of energy  $E$  the phase then runs with  $ET_0/T$ , i.e. at the local clock rate, and the amplitude follows the local energy density, not a probability (Doc. 230). This is the gravitational redshift (Doc. 078): gravitation is already contained in the equation; beyond general relativity this form predicts nothing. No additional field and no graviton is needed for this:  $T = 1/m$  is the dual reading of the mass, and gravitation shows itself as an effect in the redshift, read as curvature of space, as a change of mass (Doc. 078) or as fractal path lengthening (Doc. 041).

## Higgs Field Coupling

The Higgs field couples to the time field through:

$$\mathcal{L}_{\text{Higgs-T}} = |D_{\text{Higgs-T}}|^2 - V(T(x, t), \Phi) \quad (17)$$

where:

$$D_{\text{Higgs-T}} = T(x, t)(\partial_\mu + igA_\mu)\Phi + \Phi\partial_\mu T(x, t) \quad (18)$$

This establishes the fundamental connection:

$$T(x, t) = \frac{1}{y\langle\Phi\rangle} \quad (19)$$

## .4 Matter Field Coupling Through Conformal Transformations

### Conformal Coupling Principle

All matter fields couple to the time field through conformal transformations of the metric:

$$g_{\mu\nu} \rightarrow \Omega^2(T(x, t))g_{\mu\nu}, \quad \text{where} \quad \Omega(T(x, t)) = \frac{T_0}{T(x, t)} \quad (20)$$

**Dimensional verification:**  $[\Omega(T(x, t))] = [T_0/T(x, t)] = [E^{-1}]/[E^{-1}] = [1]$  (dimensionless) ✓

### Scalar Field Lagrangian

For scalar fields:

$$\mathcal{L}_\phi = \sqrt{-g}\Omega^4(T(x, t))\left(\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2\right) \quad (21)$$

**Dimensional verification:**

- $[\Omega^4(T(x, t))] = [1]$  (dimensionless)
- $[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi] = [E^2][E^2] = [E^4]$
- $[m^2\phi^2] = [E^2][E^2] = [E^4]$
- Total:  $[E^{-4}][1][E^4] = [E^0]$  (dimensionless) ✓

### Fermion Field Lagrangian

For fermion fields:

$$\mathcal{L}_\psi = \sqrt{-g}\Omega^4(T(x, t))\left(i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi\right) \quad (22)$$

**Dimensional verification:**

- $[i\bar{\psi}\gamma^\mu\partial_\mu\psi] = [E^{3/2}][1][E][E^{3/2}] = [E^4]$

- $[m\bar{\psi}\psi] = [E][E^{3/2}][E^{3/2}] = [E^4]$
- Total:  $[E^{-4}][1][E^4] = [E^0]$  (dimensionless) ✓

## .5 Connection to Higgs Physics and Parameter Derivation

### The Universal Scale Parameter from Higgs Physics

The T0 model's fundamental scale parameter,  $\xi = 4/30000$ , can be checked against Higgs physics. The calculation yields:

$$\xi \approx \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.30 \times 10^{-4} \quad (23)$$

where:

- $\lambda_h = m_h^2/(2v^2) \approx 0.129$  (Higgs self-coupling, dimensionless)
- $v \approx 246$  GeV (Higgs VEV, dimension  $[E]$ )
- $m_h \approx 125$  GeV (Higgs mass, dimension  $[E]$ )

**Complete dimensional verification:**

$$[\xi] = \frac{[1][E^2]}{[1][E^2]} = \frac{[E^2]}{[E^2]} = [1] \quad (\text{dimensionless}) \checkmark \quad (24)$$

#### Universal Scale Parameter

**Key Insight:** The parameter  $\xi(m) = 2Gm/\ell_P$  scales with mass, revealing the **fundamental unity of geometry and mass**. At the Higgs mass scale,  $\xi_0 \approx 1.33 \times 10^{-4}$  provides the natural reference value that characterizes the coupling strength between the time field and physical processes in the T0 model.

### Connection to $\beta_T$ Parameter

The relationship between the scale parameter and the time field coupling is established through:

$$\beta_T = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2 \xi} = 1 \quad (25)$$

This relationship, combined with the condition  $\beta_T = 1$  in natural units, links  $\xi$  to the Higgs sector; with  $\xi = 4/30000$  and the SM values it gives  $\beta_T \approx 0.976$ , i.e. consistency to about 2.4 %.

### Geometric Modifications for Different Field Regimes

The universal scale parameter  $\xi$  undergoes geometric modifications depending on the field configuration:

- **Localized fields:**  $\xi = 1.33 \times 10^{-4}$  (full value)
- **Infinite homogeneous fields:**  $\xi_{\text{eff}} = \xi/2 = 6.7 \times 10^{-5}$  (cosmic screening)

This factor of 1/2 reduction arises from the  $\Lambda_T$  term in the modified field equation for infinite systems and represents a fundamental geometric effect rather than an adjustable parameter.

## .6 Complete Total Lagrangian Density

### Full T0 Model Lagrangian

The complete Lagrangian density for the T0 model is:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{time}} + \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\phi} + \mathcal{L}_{\psi} + \mathcal{L}_{\text{Higgs-T}} \quad (26)$$

where the components are:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[ \frac{1}{2} g^{\mu\nu} \partial_{\mu} T(x, t) \partial_{\nu} T(x, t) - V(T(x, t)) \right] \quad (27)$$

$$\mathcal{L}_{\text{gauge}} = \sqrt{-g} \left( -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \quad (28)$$

$$\mathcal{L}_{\phi} = \sqrt{-g} \Omega^4(T(x, t)) \left( \frac{1}{2} g^{\mu\nu} \partial_{\mu} \phi \partial_{\nu} \phi - \frac{1}{2} m^2 \phi^2 \right) \quad (29)$$

$$\mathcal{L}_{\psi} = \sqrt{-g} \Omega^4(T(x, t)) \left( i \bar{\psi} \gamma^{\mu} \partial_{\mu} \psi - m \bar{\psi} \psi \right) \quad (30)$$

$$\mathcal{L}_{\text{Higgs-T}} = \sqrt{-g} |D_{\text{Higgs-T}}|^2 - V(T(x, t), \Phi) \quad (31)$$

**Dimensional consistency:**  $\mathcal{L}_{\phi}$  and  $\mathcal{L}_{\psi}$  have dimension  $[E^0]$  (dimensionless);  $\mathcal{L}_{\text{time}}$  is not uniform with the assignments of Section .3 (kinetic term  $[E^{-2}]$ , potential term  $[E^0]$ ).

## .7 Cosmological Applications

### Modified Gravitational Potential

The T0 model predicts a modified gravitational potential:

$$\Phi(r) = -\frac{GM}{r} + \kappa r \quad (32)$$

where  $\kappa$  depends on the field geometry:

- **Localized systems:**  $\kappa = \alpha_{\kappa} H_0 \xi$
- **Cosmic systems:**  $\kappa = H_0$  (Hubble constant)

### Static Universe Interpretation

The T0 model explains cosmological observations without spatial expansion:

- **Redshift:** Energy loss to time field gradients
- **Cosmic microwave background:** Equilibrium radiation in static universe
- **Structure formation:** Gravitational instability with modified potential
- **Dark energy:** Emergent from  $\Lambda_T$  term in field equation

## .8 Experimental Predictions and Tests

### Distinctive T0 Signatures

The T0 model makes specific testable predictions using the universal scale parameter  $\xi \approx 1.33 \times 10^{-4}$ :

### 1. Modified gravitational dynamics:

$$v^2(r) = \frac{GM}{r} + \kappa r^2 \quad (33)$$

### 2. Energy-dependent quantum effects:

$$\Delta t = \frac{\xi}{c} \left( \frac{1}{E_1} - \frac{1}{E_2} \right) \frac{2Gm}{r} \quad (34)$$

A wavelength-dependent redshift does not hold, because the cosmological redshift is achromatic: the geometric path lengthening stretches all wavelengths by the same factor. A measurable  $\xi$  addition to the anomalous magnetic moment does not exist either (Section .10).

## Precision Tests

The fixed-parameter nature allows stringent tests:

- **No free parameters:** All coefficients derived from  $\xi \approx 1.33 \times 10^{-4}$
- **Cross-correlation:** Same parameters predict multiple phenomena
- **Universal predictions:** Same  $\xi$  value applies across all physical processes
- **Quantum-gravitational connection:** Tests of unified framework

## .9 Dimensional Consistency Verification

### Complete Verification Table

| Equation                                                    | Left Side               | Right Side                                      | Status |
|-------------------------------------------------------------|-------------------------|-------------------------------------------------|--------|
| Time field definition                                       | $[T] = [E^{-1}]$        | $[1/\max(m, \omega)] = [E^{-1}]$                | ✓      |
| Field equation                                              | $[\nabla^2 m] = [E^3]$  | $[4\pi G \rho m] = [E^3]$                       | ✓      |
| $\beta$ parameter                                           | $[\beta] = [1]$         | $[2Gm/r] = [1]$                                 | ✓      |
| $\xi$ parameter (Higgs)                                     | $[\xi] = [1]$           | $[\lambda_h^2 v^2 / (16\pi^3 m_h^2)] = [1]$     | ✓      |
| $\beta_T$ relationship                                      | $[\beta_T] = [1]$       | $[\lambda_h^2 v^2 / (16\pi^3 m_h^2 \xi)] = [1]$ | ✓      |
| Energy loss rate                                            | $[dE/dr] = [E^2]$       | $[g_T \omega^2 2G/r^2] = [E^2]$                 | ✓      |
| Modified potential                                          | $[\Phi] = [E]$          | $[GM/r + \kappa r] = [E]$                       | ✓      |
| Lagrangian density ( $\mathcal{L}_\phi, \mathcal{L}_\psi$ ) | $[\mathcal{L}] = [E^0]$ | $[\sqrt{-g} \times \text{density}] = [E^0]$     | ✓      |
| QED correction                                              | $[a_t^{(T0)}] = [1]$    | $[\alpha \xi^2 / 2\pi] = [1]$                   | ✓      |

**Table 1:** Complete dimensional consistency verification for T0 model equations

## .10 Connection to Quantum Field Theory

### Modified Dirac Equation

The Dirac equation in the T0 framework becomes:

$$[i\gamma^\mu(\partial_\mu + \Gamma_\mu^{(T)}) - m(x, t)]\psi = 0 \quad (35)$$

where the time field connection is:

$$\Gamma_\mu^{(T)} = \frac{1}{T(x, t)} \partial_\mu T(x, t) = -\frac{\partial_\mu m}{m} \quad (36)$$

### QED Correction with Universal Scale: Excluded by Measurement

A time-field addition to the anomalous magnetic moment with the universal scale parameter would have the form:

$$a_e^{(T0)} = \frac{\alpha}{2\pi} \cdot \xi^2 \cdot I_{\text{loop}}, \quad I_{\text{loop}} = \frac{1}{12} \quad (37)$$

With the physical coupling this gives  $\alpha/(2\pi) \cdot \xi^2/12 = 1.7 \times 10^{-12}$ ; the value  $2.34 \times 10^{-10}$  arises only with  $1/(2\pi)$ , i.e. with  $\alpha = 1$ . Both values are excluded: measurement minus QED calculation gives  $+3.4(1.6) \times 10^{-13}$  (Rb) or  $-10.1(2.7) \times 10^{-13}$  (Cs), depending on the determination of  $\alpha$ ; an addition of  $1.7 \times 10^{-12}$  would lie  $8.6\sigma$  or  $10\sigma$  away. A measurable  $\xi$  addition to  $a_e$  does not exist.